

DES001

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THE FINBIN EXHIBITION

Making Recycling Fun!

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SPONGE PHASE

PHASE 1 + PHASE 2

PROJECT BRIEF

The Issue:

Dying ecosystems are no stranger to our modern-day society. Melting glaciers, acidifying oceans, and bleaching corals, one problem atop another connect to form a downward spiral of systemic decay. From one issue to another, the root of all ocean-related issues has always come down to one central factor: our human influence. Even with wide-scale efforts in going green today, cutting down on large-scale waste-production factors such as greenhouse gas emissions from vehicles seems like a distant fairytale thanks to the dishonest corporate influence guiding the general crowd towards a falsified green. Instead, many have gathered amongst the general public in hopes of achieving a cut in human influence in the form of recycling. Yet, despite such widespread knowledge of the importance of recycling, engagement seems to remain bottlenecked at a relatively low rate throughout the years. The truth is, as humans, we simply aren't accustomed to prioritizing goals that are irrelevant to the benefit of our daily lives. Remember the last time you learned about an interesting topic in a class or seminar, and remind yourself how long it was before you completely forgot about that topic. Everyone understands the importance, it just never takes priority. This is why on top of spreading awareness, a product design that makes recycling appealing and enjoyable is crucial to its overall success.

What Our Project Does:

Our product not only contributes to the ongoing effort of encouraging recycling but also attracts individuals who would otherwise not have been the target audience through utilizing eye-catching designs for initial appeal as well as a small reward system for future longevity. Traditional recycling methods have repeatedly proven unsuccessful at engaging the public effectively, but a fun twist on trash cans may be the antidote to this long-running struggle for engagement.

Inspiration and Design:

A working case of disposal systems that reward users already exists in Germany; in fact, "*A look at Germany's bottle deposit scheme*" reveals that the system is one of the primary reasons that Germany "boasts a return rate of above 98%" (Ruiz et al.). A charge known as a "pfand" is charged when a consumer purchases a drink to ensure that bottles are properly recycled. Similar bottle bills were passed in up to ten states in the US, yet their limited accessibility meant that their efficiency remains unideal compared to the practicality of systems offered by Germany. Thus, rather than creating specific stations that are limited in availability, our project aims to create replacement lids for existing trash cans that can interact with users and offer rewards. Users may either scan their own QR code for immediate rewards or redeem these rewards as points that allow them to redeem digital vouchers at a reduced price. Dedicated or long-time users of the service may become affiliated and earn bonuses or even get paid as a job. The funding for this project can be achieved through government funds obtained through current bottle bills.

Sources:

Maureen, Veronica. "Recycling Incentive Scheme." Plastic Smart Cities, 6 Aug. 2023, plasticsmartcities.org/recycling-incentive-scheme/.

LOOK BOOK

Inspirations based on pre-existing designs as well as previously successful case samples



"Study-in-Germany Pfand System in Germany -." In, www.study-in-germany.de/en/community/pfand-system-in-germany/. Accessed 7 Oct. 2024.



Eident, Kathryn. "Your Glass Bottle Could End up in a Cape Cod Road - and That's a Good Thing." CAI, CAI, 13 Feb. 2020, www.capeandislands.org/news/2019-05-14/your-glass-bottle-could-end-up-in-a-cape-cod-road-and-thats-a-good-thing.



"Trash Can Redesign." Trash Can Redesign | Public Works, sfpublicworks.org/trashcanredesign. Accessed 7 Oct. 2024.

TARGET AUDIENCE



BROCK

AGE: 36

JOB: Unemployed

Needs quick ways to be affiliated and paid



EDGAR

AGE: 12

JOB: Student

Looking to earn some pocket money, curious to learn more



BEATRICE

AGE: 76

JOB: Retired

Has a lot of spare time, willing to give back to the community



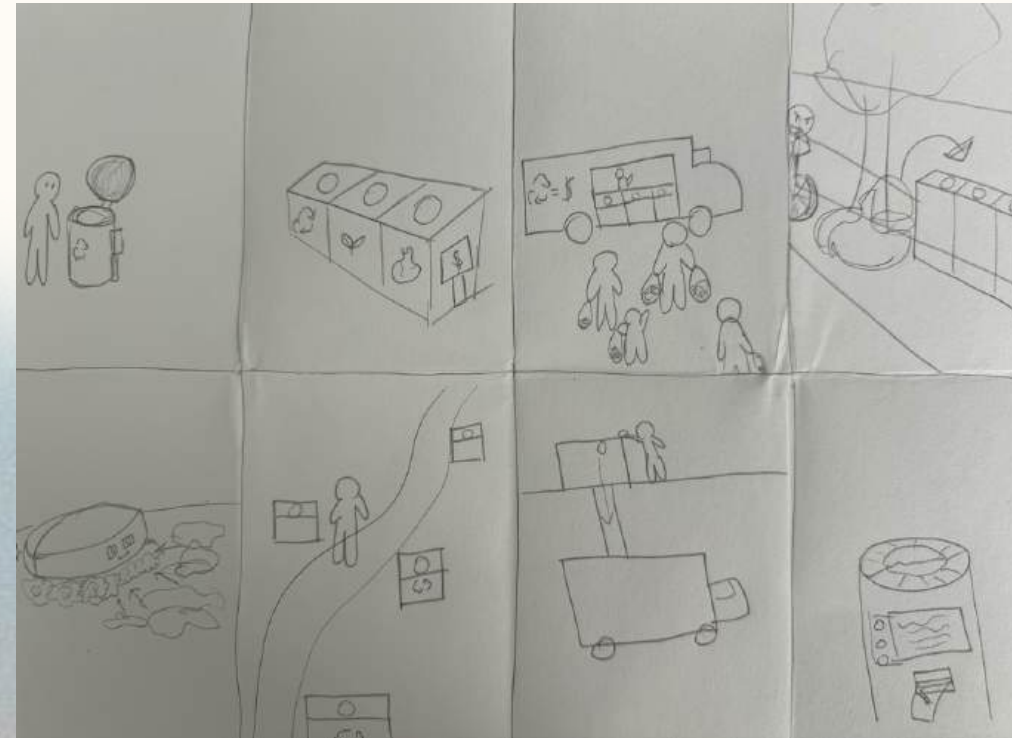
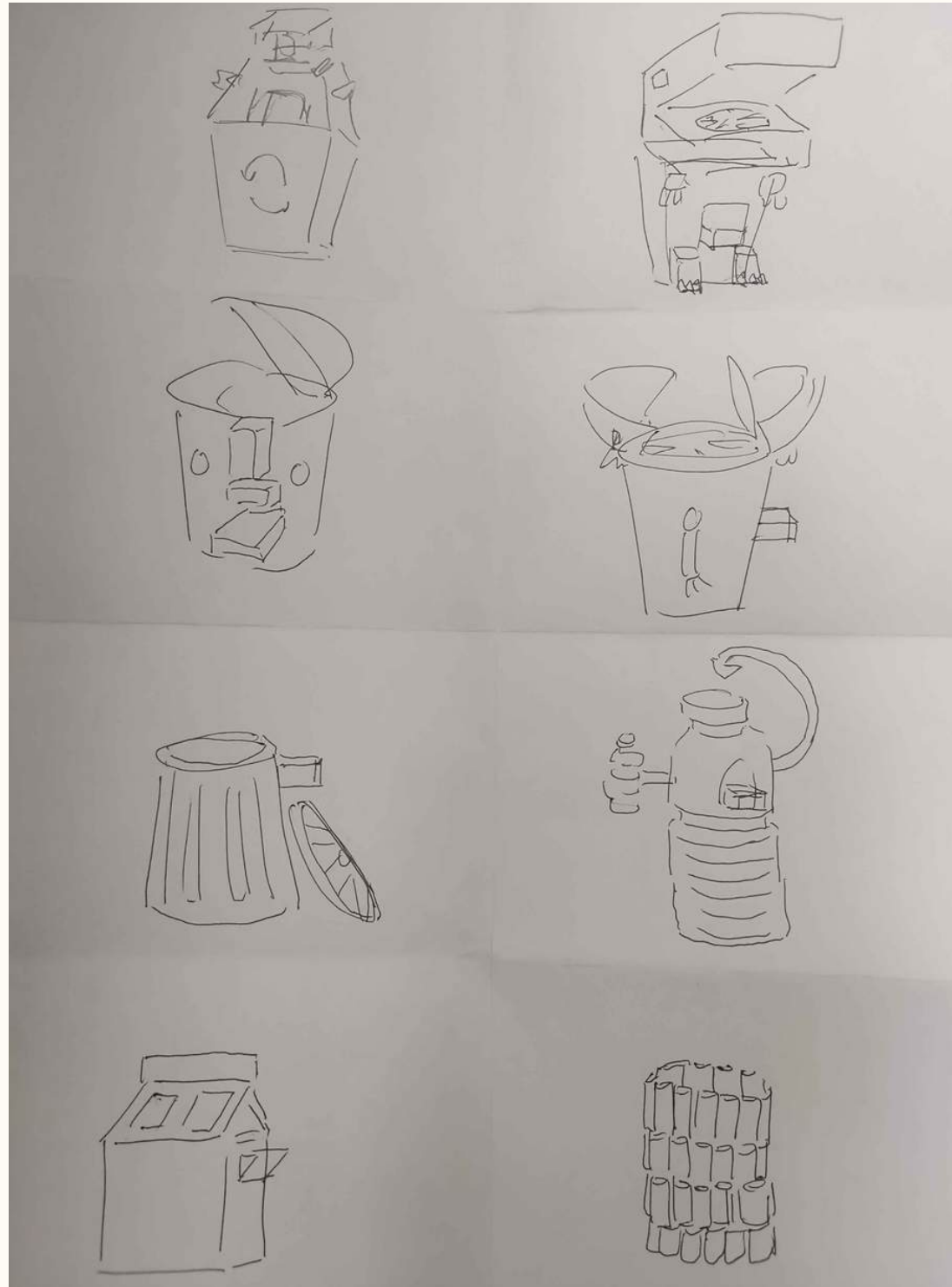
HUNTER

AGE: 24

JOB: Student / Part Time

Tight on both money and time

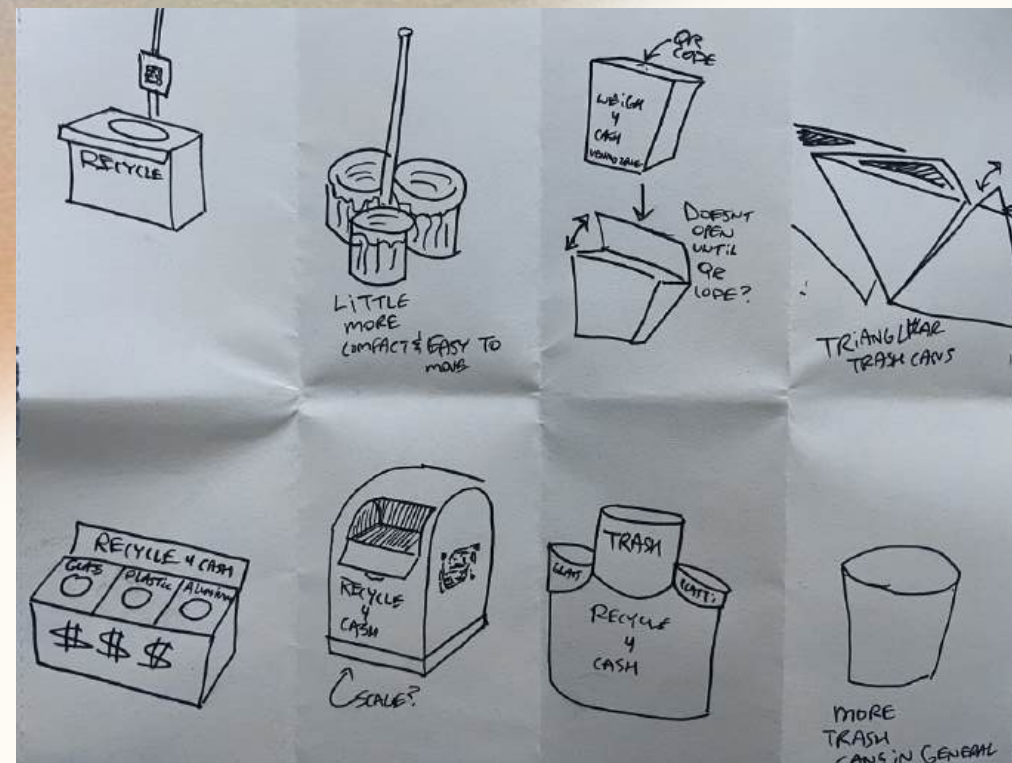
CRAZY 8



Credit: Kevyn S. Campos



Credit: Kyra Calderhead



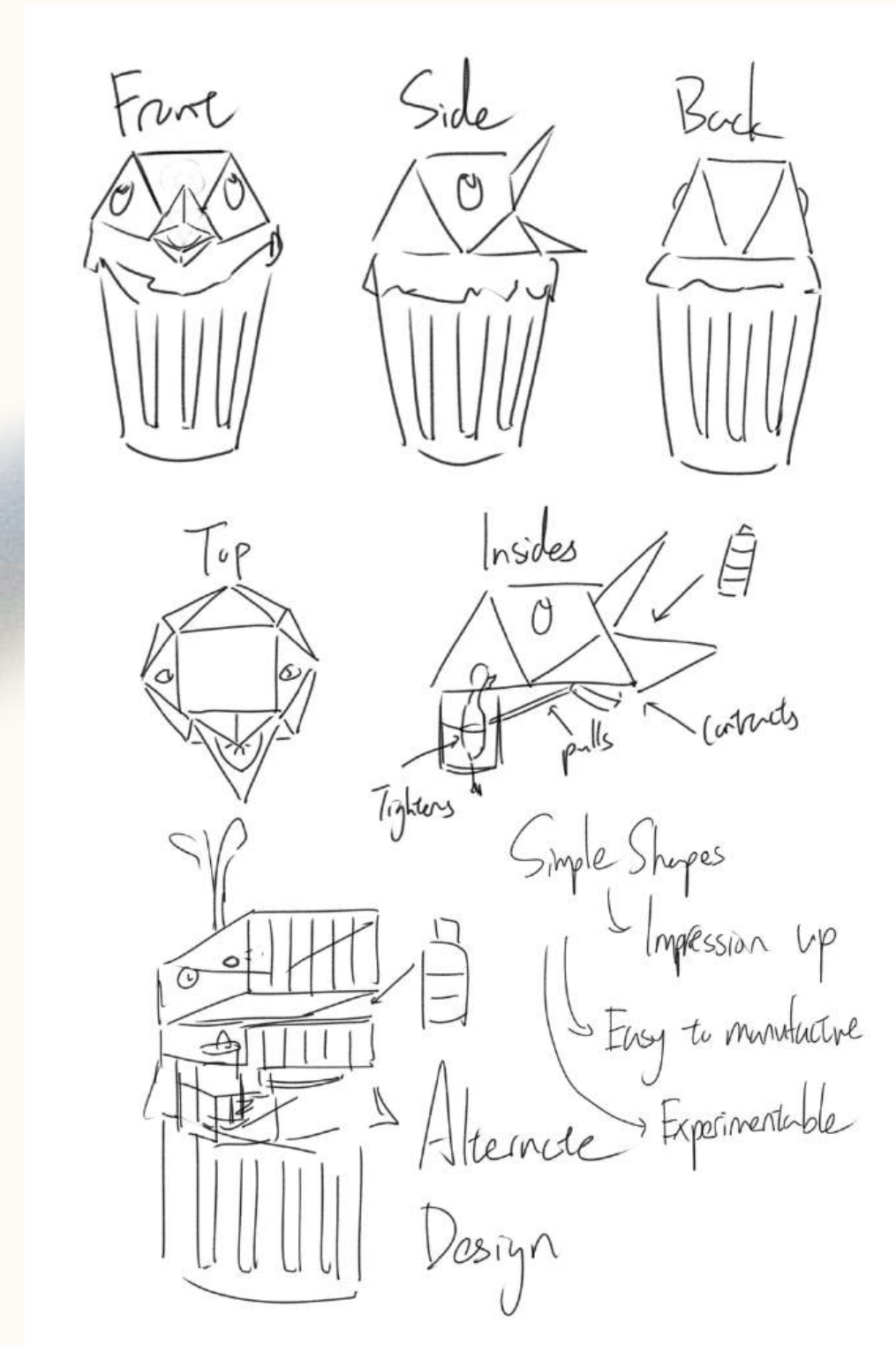
Credit: Feliks Karapetyan

Upon discussion, our group decided that building an interactive trash can that appealed to passersby would be the route that we chose for the final project. This came in multiple forms, including modifying the trash can, bundling them in bundles, or simply increasing the number of trash cans in the area. We ultimately landed on designing animal-themed trash cans that featured interactive elements as our main direction.

SKETCH SOLUTION



Credit: Feliks Karapetyan



DES001

FILTER PHASE

DES001

VOICE PHASE

PHASE 1

KANBAN BOARD

Filter I | 1 ...

Kyra | 5 ...

Feliks | 3 ...

In progress | 1 ...

Brand Identity Design
In progress

Done | 13 ...

Moodboard & Analysis (Individual)
Done

KanBan Board
Done

Contribution Report 1
Done

User Interaction Sketch
Done

Cardboard Collecting
In progress

Moodboard & Analysis (Individual)
Done

Empathy Maps (2)
In progress

Kevin | 4 ...

Contribution Report 4

Kevyn | 3 ...

BOM (Bill of Materials)
Done

Moodboard & Analysis (Individual)
Done

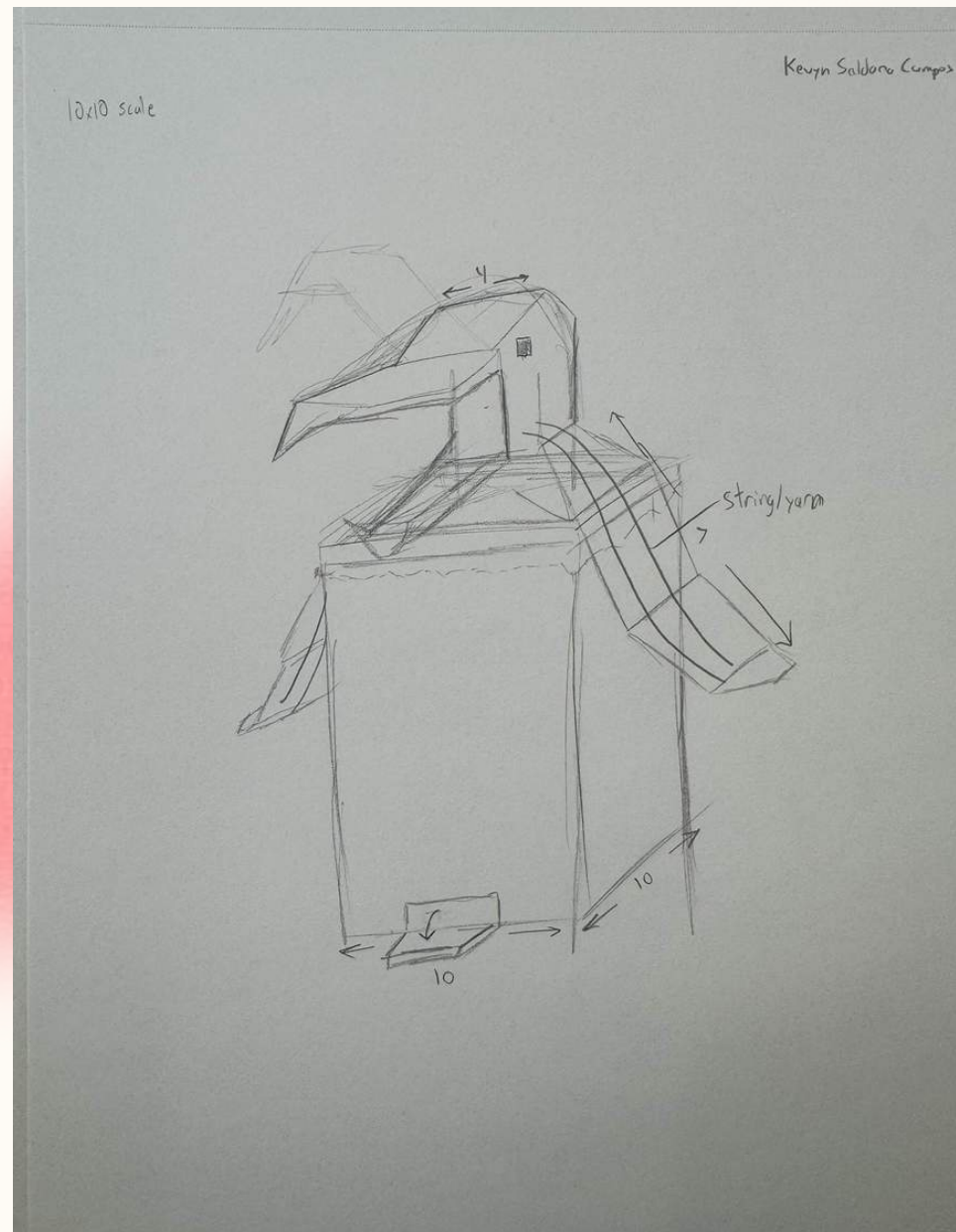
Perspective Sketch (choose what perspective based on canvas options)
In progress

Moodboard & Analysis (Individual)
Done

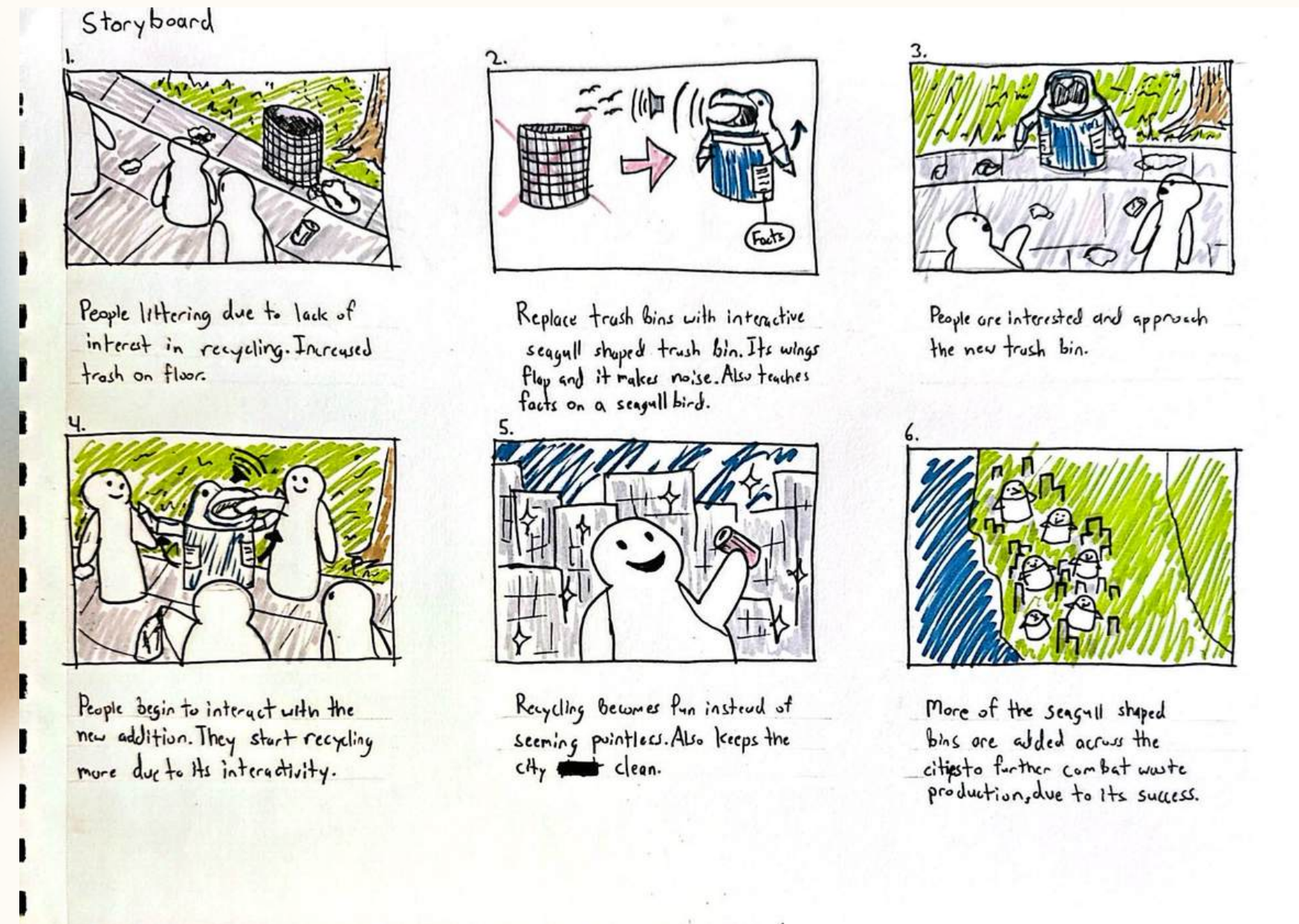
Contribution Report 2
Done

Storyboard
Done

DESIGN SKETCHES



Credit: Kevyn S. Campos
SCALE DIAGRAM

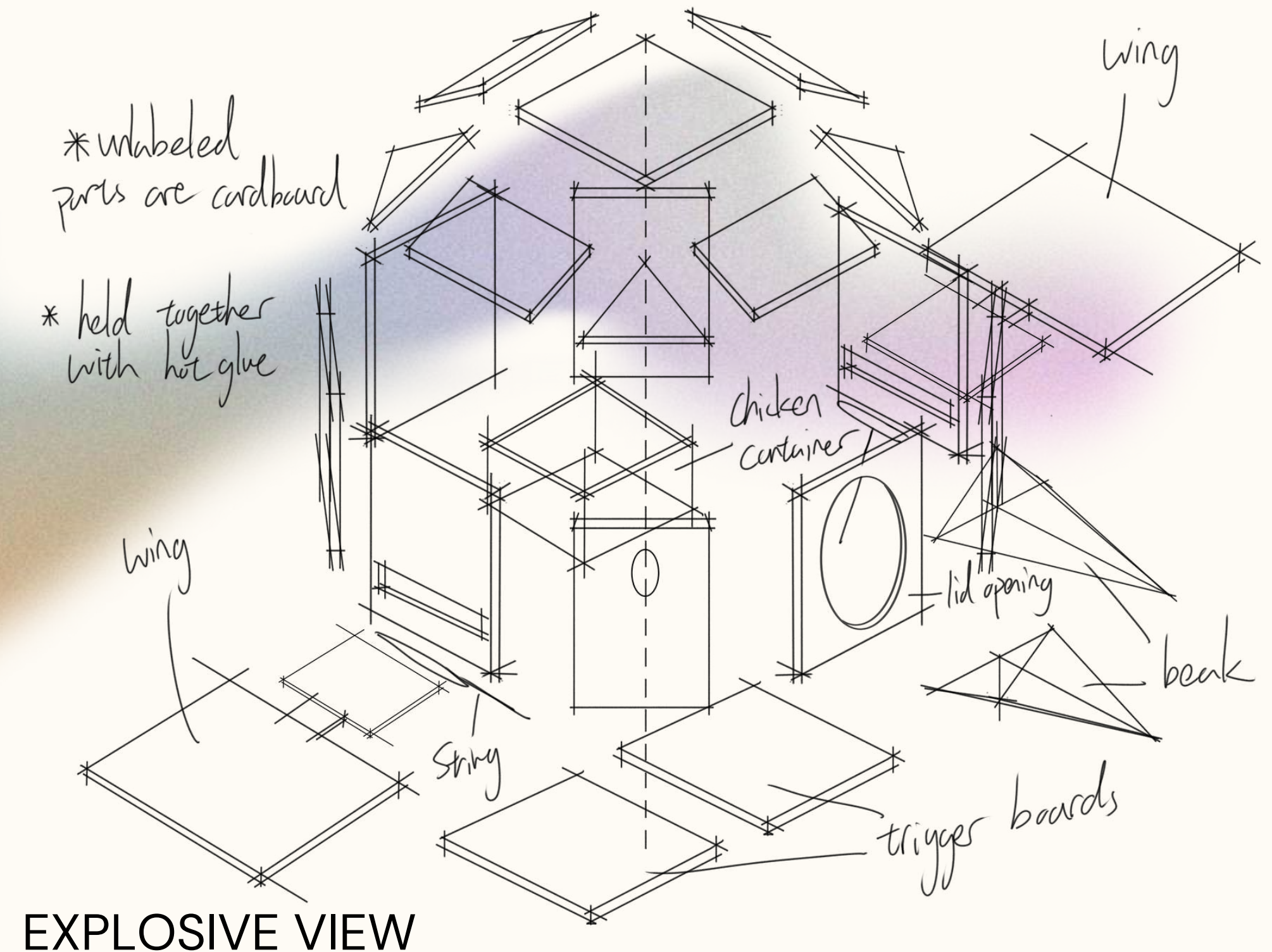


Credit: Kevyn S. Campos
STORYBOARD

DESIGN SKETCHES



Credit: Kyra Calderhead
USER INTERFACE

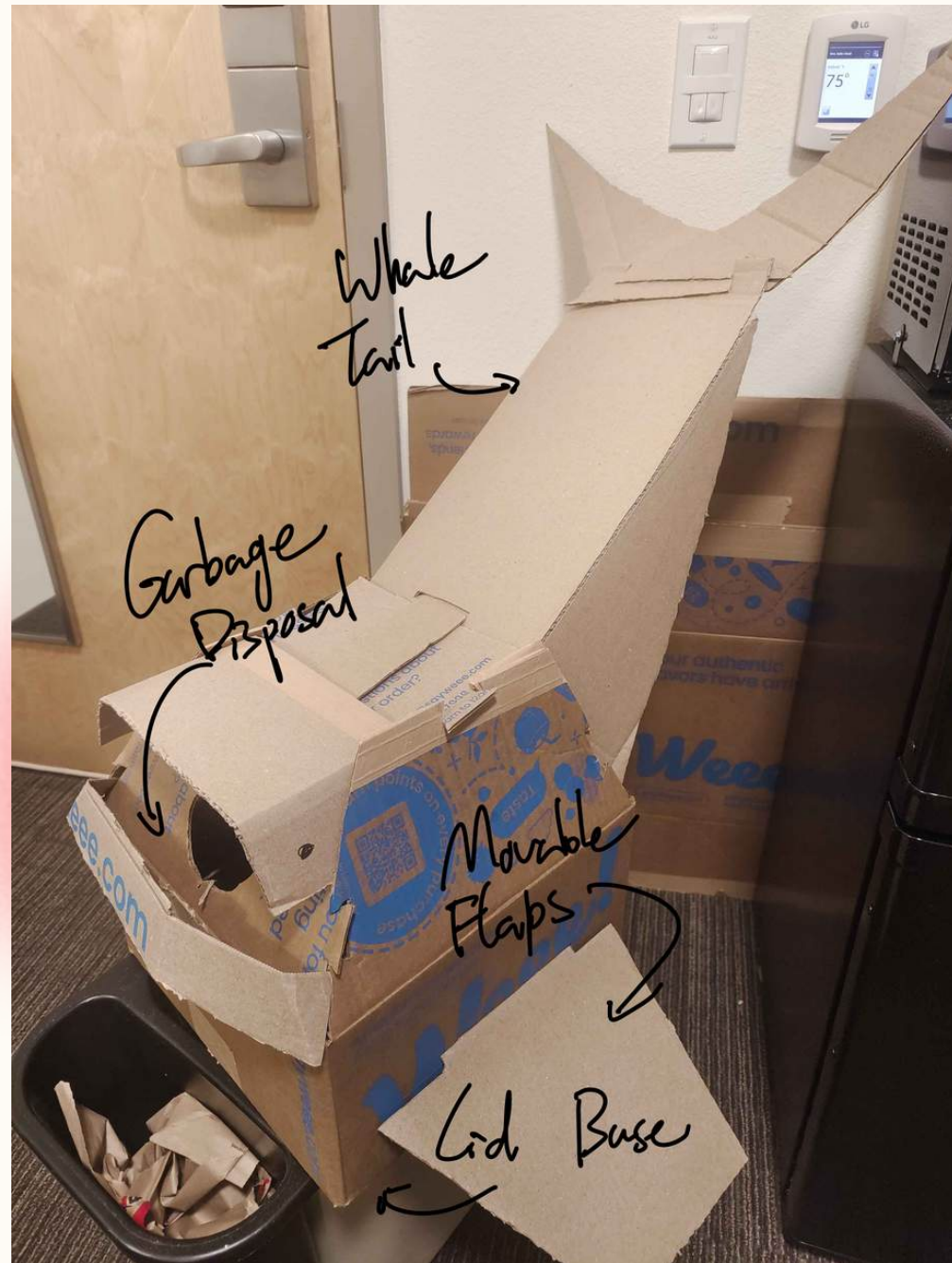


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VOICE PHASE

PHASE 2

RAPID PROTOTYPING



PROTOTYPE: WHALE

A core feature about prototype two is the omission of adhesives for the hinges and fixed edges. Using cardboard slits gives the design a cohesive feel without having an access to distracting elements that take away from the character. Furthermore, the change from the seagull to the whale is significant in the development of our design as it opened up a new discovery: simplification and ensuring attractive design. Through switching to the whale design, we omitted the issue of the beak being possibly too “sharp” or too “enclosed” to feel inviting, especially for younger children who we intend to target.

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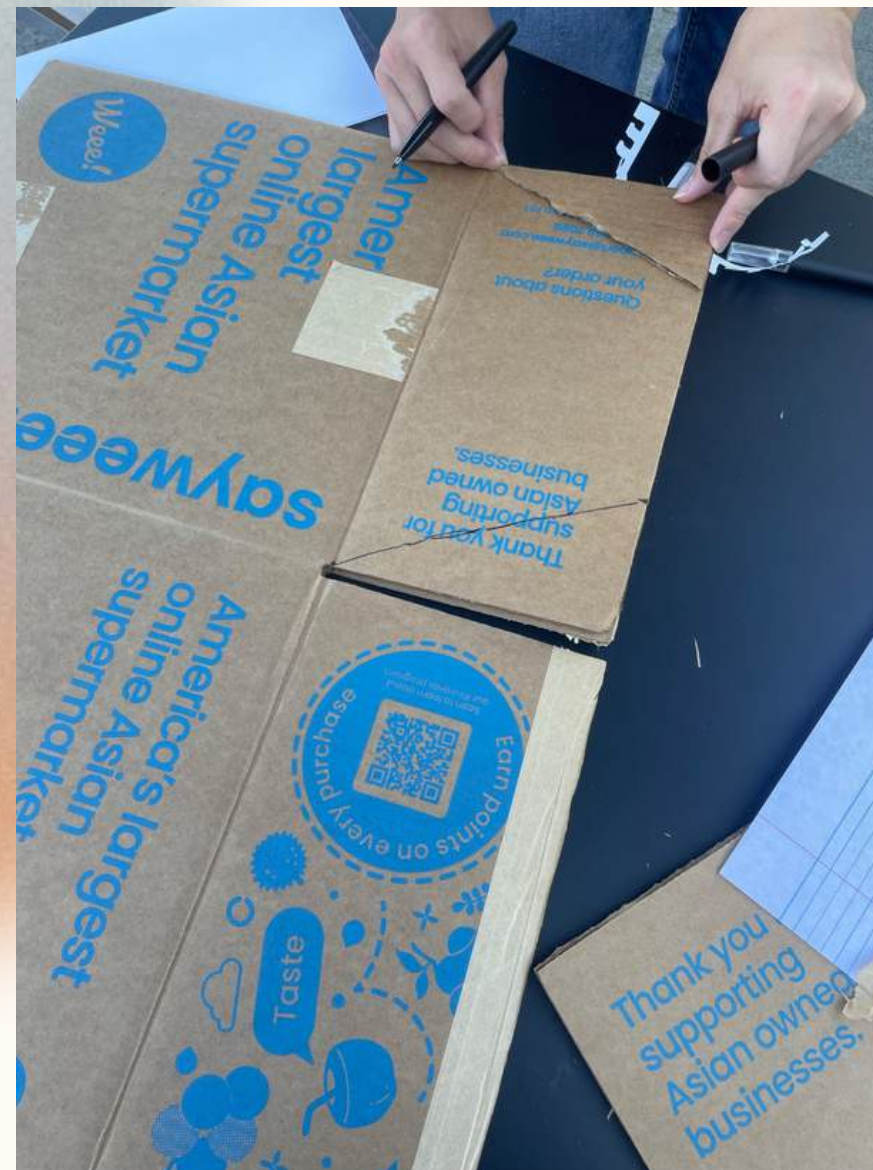
VOICE PHASE

PHASE 3

USER STUDY REPORT

- A list of the most common positive feedback (at least 3)
 - Design that is inclusive to all ages
 - Creative way for people to want to recycle
 - Many aesthetic avenues to pursue with different marine animals
 - Addresses coastal environmental concerns through enforcing recycling behavior rather than implementing a new action for people to take
- A list of the most common constructive suggestions (at least 3)
 - Making the design colorful and more playful
 - Mechanism to keep insects out rather than an open hole
 - Larger hole or mouth opening so people can throw recyclable in quickly without the barrier of time preventing them from using the product
- Describe your plan to improve the prototype based on the feedback
 - To improve the prototype based on peer feedback, the group plans on implementing the whale “floss” behind their mouth to help filter out insects without slowing down the process of recycling
 - Further, the group plans to use slotting methods to provide more structural integrity and account for environmental stressors on the product.
 - The slot design also adds a reproducibility factor to the lid, making it a kit that could be easily reproduced for home DIY projects, thus adding an educational factor to the design.
 - Further, based on many comments concerning the color and texture of the product, we plan on purchasing spray paint to coat the design without saturating it / weighing it down with too much paint. This will help add visual appeal in addition to ensuring the product reads as a whale and relates to coastal community concerns.

BODY



Materials used: Cardboard, cardboard cutter or blade, marker

Photographs: Kyra Calderhead

Descriptions: The body starts its process with a regular square recyclable box. This box is opened and 45 degree cuts are made into the top folding pieces that close the box. These cuts are important to be at this angle due to the shape and creation of the body, as well as being able to fit the head on properly. If these slots are not done correctly, the head of the whale will simply not connect as efficiently as well as creating problems for trash getting in.

BODY



Materials used: Cardboard, cardboard cutter or blade, scissors, marker

Photographs: Kyra Calderhead

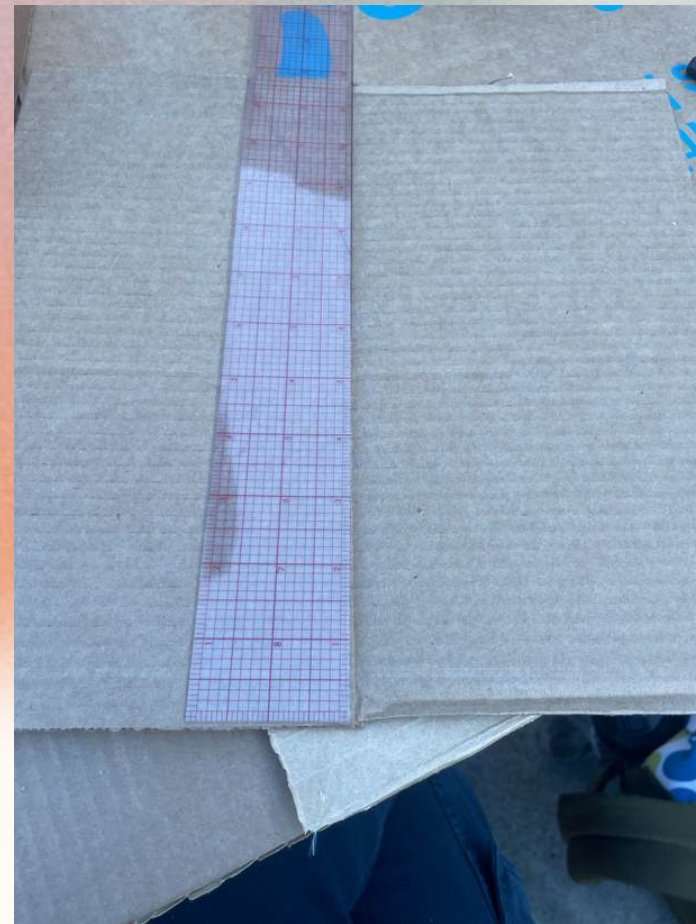
Descriptions: The top flaps of the cardboard box begin to form the neck and resting area for the head to be placed. In a factory, this can be replicated by cutting the cardboard into trapezoids with 45 degree angles, once the trapezoids are complete, sections of the side lengths are formed into slots and slot receivers to allow for the neck of the whale to be present. This is also the area where trash will slide through to reach the bottom of the trash can, which needs to be created using trapezoids in order to form the largest diameter/area for bigger bottles such as gatorade to be recyclable in our design.

WINGS

Materials used: Cardboard, cardboard cutter or blade, ruler

Photographs: Kyra Calderhead

Descriptions: The start of the wings involved a two section rectangular piece that was folded on one section. This fold is important because it will be the part of the cardboard that slides inside the body and allows for the flapping action to work. The longer and wider section of the cardboard is kept outside and simulates the wings/fins of the whale.



WINGS

Materials used: Cardboard, cardboard cutter or blade, popsicle sticks

Photographs: Kyra Calderhead

Descriptions: Once the flaps of the fins are created and slid into the body, the use of popsicle sticks allows for support as well as the flapping action to work more generously. It provides a steady support and will be included in the kit for creating our designed trashcan.



HEAD



Materials used: Cardboard, cardboard cutter or blade

Photographs: Kyra Calderhead
Descriptions: The top of the head is created using another rectangular piece of cardboard using the slots method. By cutting out and creating the slots, we have made a compact and easy to recreate method of putting the whale head and body together. This can be easily replicated by mass producing cardboard pieces with slots and slot inserts coming prepackaged and ready to assemble.

HEAD



Materials used: Cardboard, cardboard cutter or blade, popsicle sticks

Photographs: Kyra Calderhead

Descriptions: The final part of the head we are working on adding is the bottom jaw, using a filter like honeycomb cardboard, or a floss like rectangle to replicate the teeth of the whale. This in turn creates a natural filter that will stop animals from coming in and loitering around the trashcan. The top of the jaw is important to be in a 45 degree angle because that allows for the optimal angle of trash making it inside.

TAIL

Materials used: Cardboard, cardboard cutter or blade

Photographs: Kyra Calderhead

Descriptions: The tail is made by combining long rectangular boxes into the tail shape of a whale. We are working on creating a tail with a 45 degree angle that goes up similarly to the head as shown in the previous slides. The tail is an important part of the design and more pictures will be uploaded soon once the final product is implemented.

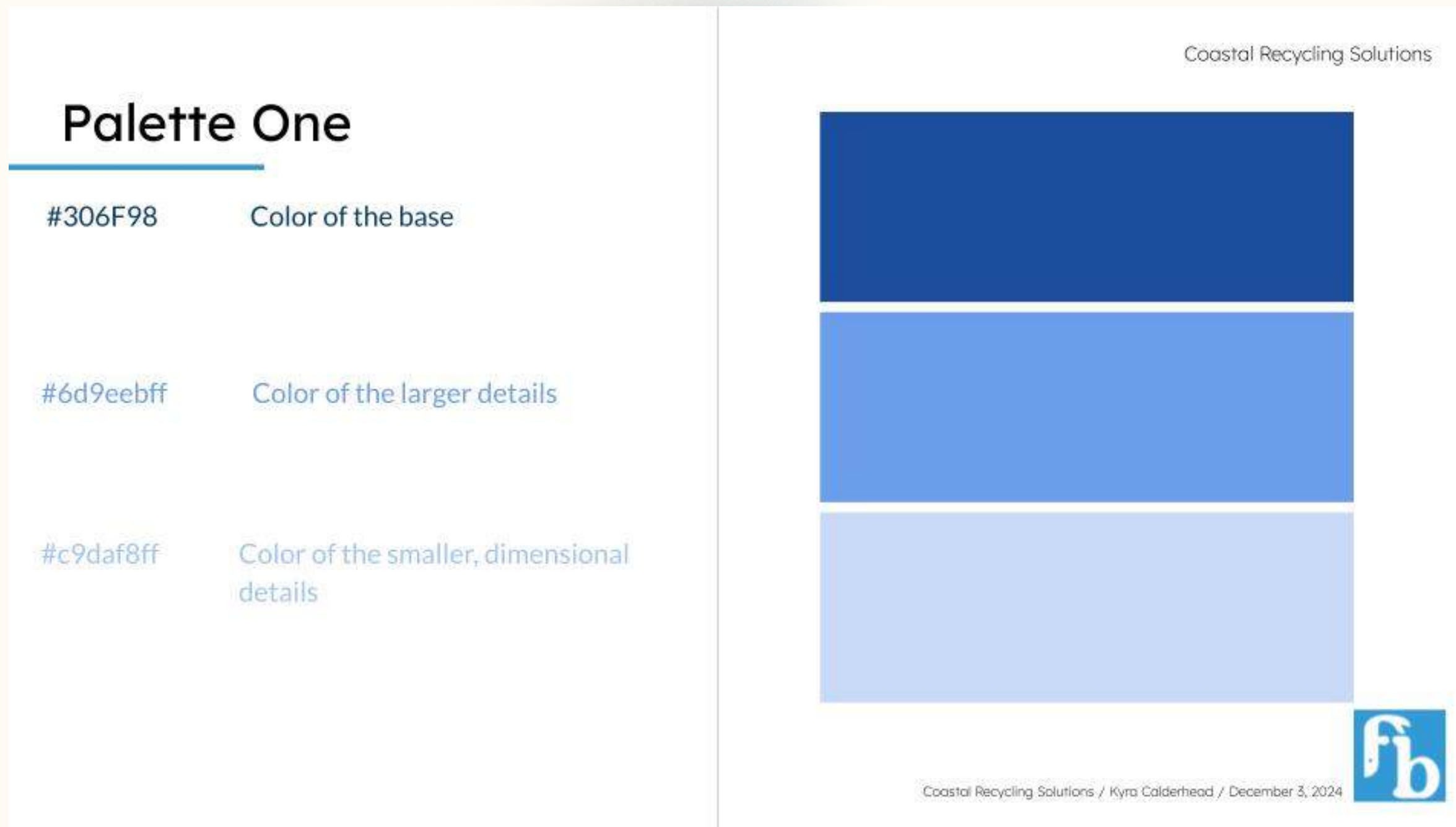


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VOICE PHASE

PHASE 4

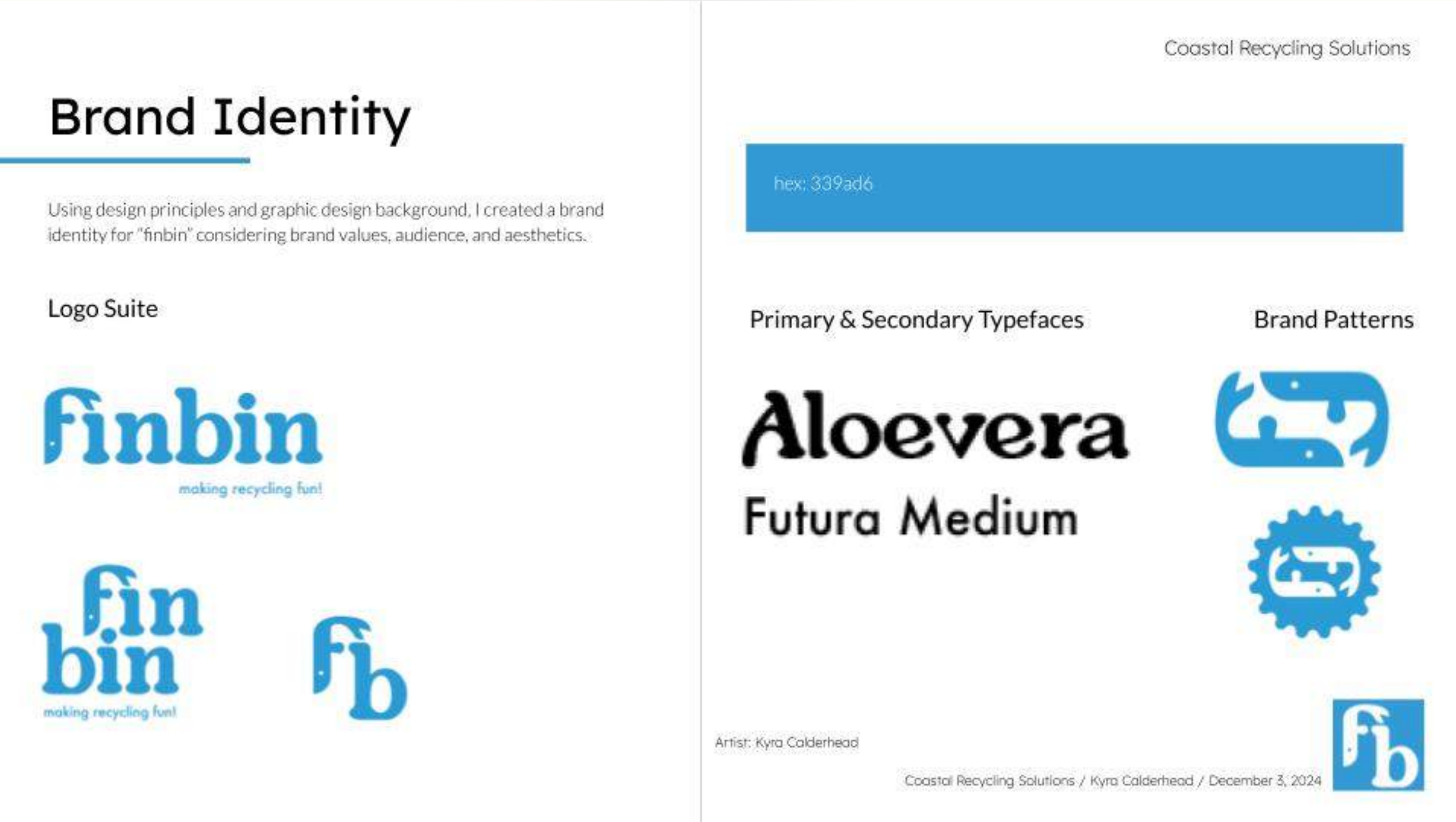
Color Palette



Kyra created the color palette that we used for the main prototype. The colors picked are meant to remind the users of the beauty of the ocean.

Credit: Kyra Calderhead

Brand Identity



Kyra also designed a logo and brand pattern for the bin, which would later be used as a core part of identifying our product.

Credit: Kyra Calderhead

Flat Design



I designed the flat design poster for our product, which will later be printed and displayed on the day of presentation. The graphics depict the core feature of our product as well as the slogan and logo of our product.

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FINAL PHASE: CHAOS IN THE COURTYARD

THE INSTALLATION



At around 8:55, our group showed up to install our prototype. With the easy-to-carry nature of our final prototype, we could easily get our prototype set up on time.

THE PRESENTATION



This image depicts a pitch that I gave to one of the people who came to check out our prototype. Here I hand off or test bottle for them to test out the prototype for themselves.

THE AUDIENCE



Here is an image of someone testing our product. Those who didn't know how our product worked prior were ecstatic to see the whale fins move. People were excited to try again with our trash can.

POST-PRESENTATION



After the presentation period, my group gets together for a group photo to commemorate the event before we wrapped up the presentation.